

Lab Assignment 2

EE5518 – AY2025/2026 Sem1

Submission Deadline: 10th November'25 11:59AM

Lab experience 1: Practicing energy-performance tradeoff in digital circuit design.

In this lab experience, you are going to build a small digital circuit and apply different design strategies for energy-performance tradeoff. Also, you need to build a TGFF circuit with the given schematic and integrate it with the digital logic as a piece of synchronous circuit.

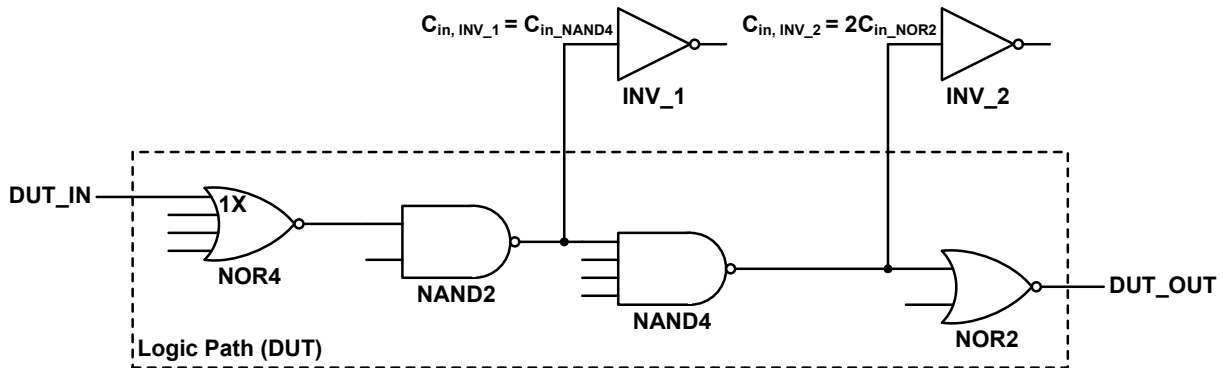


Fig. 1. Schematic of the critical path with branch paths.

Q1. The critical path of a synchronous circuit is given in Fig. 1. Draw the schematic of the logic path in gpdK045 and size the transistors for 1) minimum energy; 2) minimum delay at rising edge; 3) minimum delay at falling edge. You need to build the testbench given in Fig. 2 for delay measurements. Evaluate path delay in these sizing strategies and comment on whether it is expected or not.

Assumptions: 1) use standard cell sizing with pn ratio = 1.4;

2) VDD = 1V;

3) 1X represents the strength of a minimum-sized inverter;

4) extra inverters can be added in the path without changing the functionality.

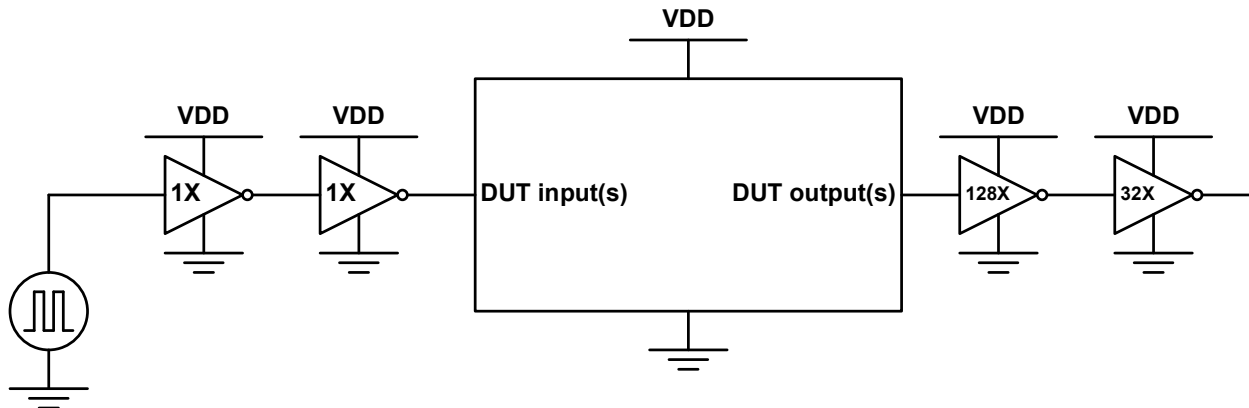


Fig 2. Testbench circuit for the measurement.

Q2. Optimize the logic path in Fig. 1 for minimum E^iD^j through determining the constant stage effort, ρ , and resizing the cells $[i,j]$ based on your matric number in Table A). Measure the energy sensitivity w.r.t. delay for the optimized design.

Table A. E^iD^j matrix assigned for each group in Lab Experience 1.

Second last character of the matriculation number of the first student in group as per Canvas	$[i,j]$
0 , 7	[2,1]
1 , 6	[1,1]
2 , 8	[1,2]
3 , 5	[1,3]
4 , 9	[3,1]

Q3. Fig. 3 gives a schematic for a Transmission Gate Flip-Flop (TGFF). It is a positive-edge triggered DFF, which takes a single input, D, to the output pin, Q, when clock, CLK, makes a low to high (rising) transition. Implement the given TGFF in 45nm and size the transistors for minimum energy (assume p_n ratio = 1.4 for inverters). Show the functionality of the circuit with schematic (pre-layout) simulations. Measure the setup time and clock-to-Q time, τ_{CQ} , of the TGFF. You need to reuse the testbench given in Fig. 2 for the measurements. Assume $V_{DD} = 1V$.

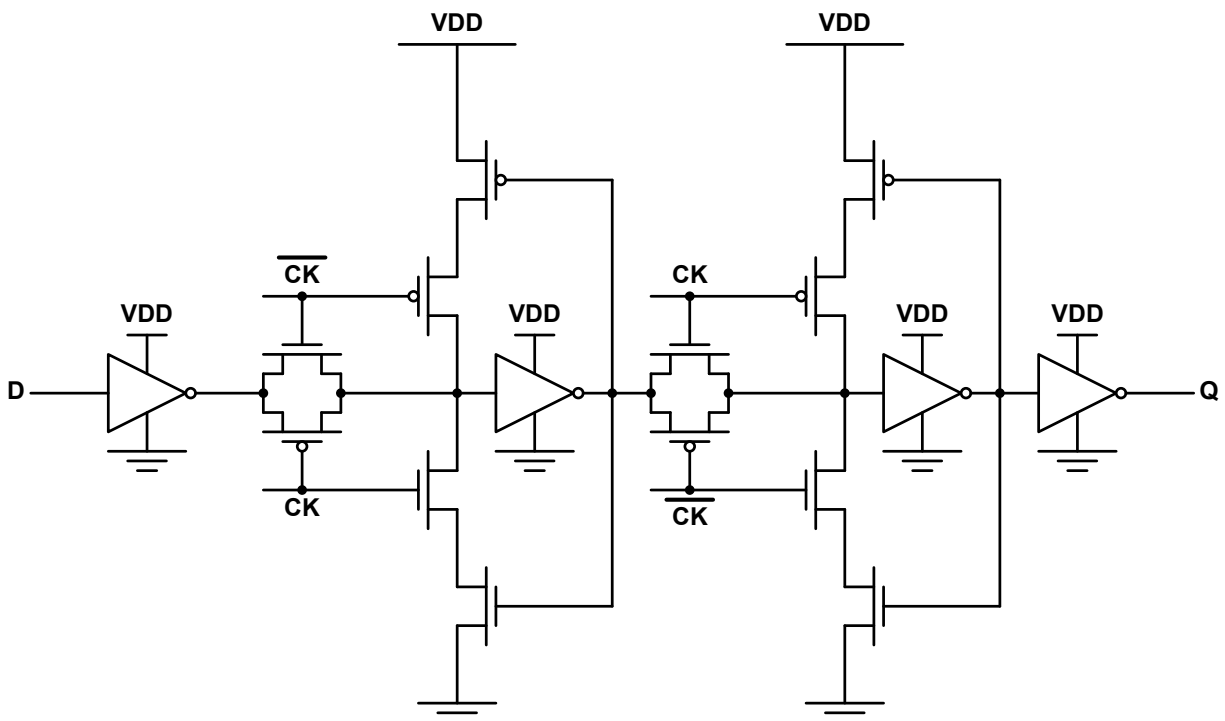


Fig. 3. Schematic of a Transmission Gate Flip-Flop (TGFF).

Q4. Assuming an ideal clock, build a synchronous circuit with two TGFFs designed in Q3 and the critical path in Q1 (minimum energy sizing). Calculate the maximum operation frequency of this synchronous circuit based on your measurements. Also, verify the maximum operation frequency through schematic (pre-layout) simulation, check the difference between calculation and simulation results, and make comments.

Lab experience 2: Studying leakage in digital circuits and wire delay with repeaters.

In this lab experience, you are going to analysis the leakage current of the logic path built in Lab Experience 1 and observe how leakage current changes while body biasing is performed. Also, you are going to practice repeater design for long wires.

Q1. Reuse the schematic of the logic path (minimum energy sizing) in Fig. 1 while ignoring the two inverters in branches. Assume $V_{DD} = 1V$. Measure the leakage current when DUT_IN is stabilized at $V_{DD}/0V$ and comment on differences between the results in the two cases. You may reuse the testbench circuit in Fig. 2 but separate the power supply for the driving and loading inverters.

Q2. Perform body biasing (from $-0.2V$ to $0.2V$ in $0.05V$ step) to the logic path built in Q1. Make a plot for leakage current vs. body biasing when DUT_IN is stable at $V_{DD}/0V$. Comment on whether these results are expected or not.

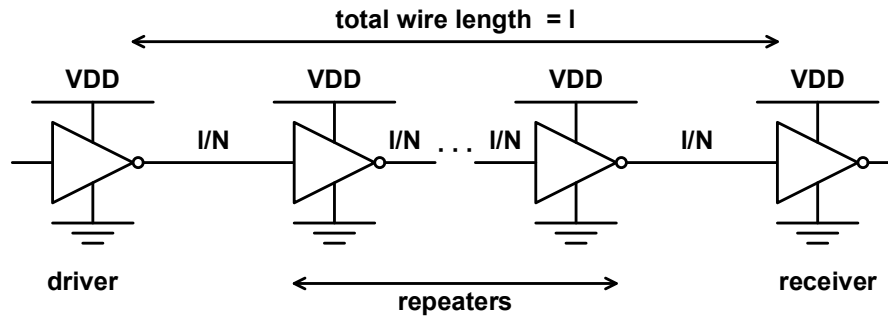


Fig. 4. Interconnect path with repeaters.

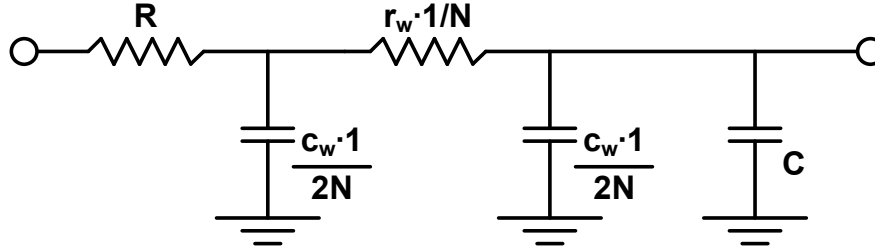


Fig. 5. Equivalent circuit for segment of repeated wire.

Q3. Show that the Elmore delay of an interconnect path with N equally repeated wire segments, and a total wire length l (Fig. 4) can be represented with below equation using π lumped RC model, then derive the equation for the best spacing between repeaters, l/N , that gives minimum Elmore delay. Assume the transistor resistance and input gate capacitance of all the drivers/repeaters are R and C respectively.

(Hint: see the equivalent circuit for segment of repeated wire in Fig. 5.)

$$\tau = N \cdot \left[R \left(\frac{c_w l}{N} + C \right) + \frac{r_w l}{N} \left(\frac{c_w l}{2N} + C \right) \right]$$

Q4. Considering the trade-off between energy-delay and l/N , optimize the repeater spacing for minimum E^D ($[i,j]$ based on your matric number in Table A).

Lab experience 3: Studying path delay and FO4 delay with voltage scaling and variations.

In this lab experience, you are going to study the relationship between a path delay and FO4 delay with voltage scaling and variations. You may assume a pn ratio of 1.4 in this lab experience and choose input transition from Table C. Perform minimum energy sizing and ignore the two inverters in branches whenever the logic path given in Fig. 1 is reused.

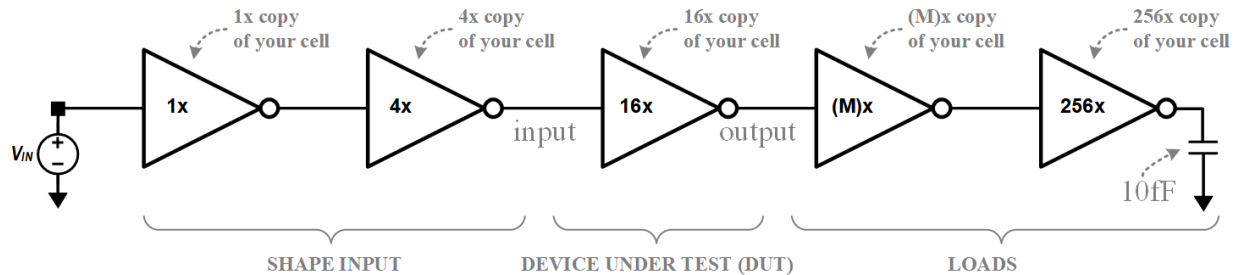


Fig. 7. Circuit template for FO4 delay measurement.

Q1. Reuse the circuit template in Fig. 7 with minimum-sized inverter cells to measure FO4 delay. Perform VDD scaling (from 0.5V to 1V in 0.05V step) for the circuit template and measure FO4 delay at these voltage points. Then, perform the same VDD scaling to the logic path given in Fig. 1 with the testbench circuits given in Fig. 2. Make a plot for VDD vs. logic path delay. Also, make another plot for VDD vs. logic path delay in terms of FO4. Comment on whether it is expected or not.

Q2. Estimate the variability of FO4 delay in gpdk045 at different corners (TT, SS, FS, SF, FF) through performing Monte Carlo simulations for 1000 runs.

Q3. Characterize the path delay in Fig. 1 at all corners and report the worst corner across P, V ($\pm 5\%$ within nominal VDD) and T (-40°C to 125°C). Verify if the ratio of the path delay and FO4 delay at the same corner is nearly constant, reporting numerical values of the ratio in a table and providing comments. Assume nominal VDD and temperature when measuring the path-delay-to-FO4-delay ratios.

Lab experience 4: Studying path-based variability.

In this lab experience, you are going to study path-based variability of the logic path in Lab Experience 1. Choose input transition from Table C. Perform minimum energy sizing and ignore the two inverters in branches whenever the logic path given in Fig. 1 is reused.

Q1. Using the Monte Carlo simulations for 1000 runs, find the maximum path delay for logic path given in Fig. 1 at typical (TT) corner such that yield is 98%.

Q2. If the maximum path delay found in Q1 is decreased by 10%, evaluate the new yield. (Hint: Using the same simulation results from Q1, determine new yield using the decreased path delay value.)

Q3. Analytically estimate the variability of the logic path (from Fig. 2) due to within-die variations under the worst-case corner (as found in Lab Experience 3 Q3) and compare the theoretical results with the Monte Carlo simulation results for 1000 runs.

Q4. Find the logic gates within the given logic path that increase the overall variability of the logic path the most.

Q5. Modify the given logic path (through sizing) with the aim to reduce the path variability by 20% from the previous value found in Q3. (Hint: New path variability = $0.8 \cdot \text{path variability found in Q3}$). Verify your calculations using Monte Carlo simulations for 1000 runs. Please report the numerical values of $(X_i \cdot \frac{\mu_i}{\mu_{\text{path}}})$.

Table C. Input transition assigned for each group in Lab Experience 3&4.

Second last character of the matriculation number of the first student in group as per Canvas	Input transition
0 - 4	High to Low
5 - 9	Low to High

Survey: individual contribution

Clearly state the contribution (%) from each group member with below template:

Matriculation number (e.g., Axxxxxxx)	Contribution (%)

Report:

- Submit your assignment in the form of a report.
- Please show all the calculation steps & relevant screenshots/figures. No marks will be given for result/value-only answers without any explanation/step.
- Clearly state any additional assumptions that you have taken.
- The final report should be in the pdf format only.
- **Clearly indicate the contribution (in %) of each student on the front page of the report. Intra-team marks may vary based on the contribution reported.**

Submission Instructions:

- **Name your file as:** Matric_Number_Assignment_2.pdf and Matric_Number_Assignment_2.zip including your own library of the design in each experience.
 - Ex. A1234567A_Assignment_2.pdf & A1234567A_Assignment_2.zip
- Upload to **Canvas EE5518 -> Files -> Assignment 2** before 10th November'25 11:59AM.
- A penalty of 25% applies for late submissions of up to 1 week. After that, the submissions will not be considered.
- **Plagiarism is penalized with a 100% penalty for all SOURCES and RECIPIENTS and will be reported for further action.**